

IOANNIS MAGOULAS

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EDUCATION

University of Maryland, Honors College

College Park, MD

Bachelor of Science, Double Degree in Computer Science & Business Info. Systems — GPA: 3.7

Expected May 2028

- Distinctions: Global Fellow, Dean's List, Honors Global Challenges and Solutions
- Relevant Coursework: Data Structures, Algorithms, Computer Systems, Linear Algebra, Discrete Structures, Statistics & Probability

EXPERIENCE

CTO Data Science Intern

June 2026 – Present

Humana

Louisville, KY

- Engineered a robust data access layer leveraging REST APIs to aggregate **Azure DevOps (ADO)** project metrics, generating automated roll-ups across epics, features, and user stories.
- Architected an end-to-end data pipeline routing aggregated ADO data into an **Azure SQL Database**, powering dynamic **Power BI** dashboards seamlessly embedded back into ADO for real-time tracking.
- Developed a custom **Claude AI Skill** that dynamically generates personalized Power BI dashboards based on natural language user prompts, accelerating data accessibility for team members.
- Pitched and delivered the platform to **Product Owners and Managers**, successfully eliminating manual spreadsheet tracking and maximizing workflow visibility through automated visual insights.

Software Developer Intern

June 2025 – Present

RealmOne

Columbia, MD

- Built and shipped full-stack features using **Java (Spring Boot)** and **Angular** on a production enterprise platform serving high-profile enterprise clients, including **Raytheon** and **Siemens**.
- Engineered **CI/CD automation** in **GitLab** with tag-based version parsing, dotenv artifact propagation, and automated RPM version management, eliminating manual release steps and reducing deployment errors.
- Developed an AI-powered chatbot with **RAG** and **Elasticsearch** to semantically search 700+ resumes, integrating a **SharePoint** webhook pipeline and deploying with **Docker** and **LiteLLM** across multiple open-source models.
- Improved API transparency and developer usability by integrating **Swagger/OpenAPI** documentation and **SPARQL** query endpoints; debugged cross-service JSON contract issues to accelerate feature integration.

Research Assistant

December 2024 – Present

University of Maryland

College Park, MD

- Contributing to “**Task Atlas**,” a framework for hierarchical task representation in AI agent planning, exploring micro-segmentation and segment embeddings for task abstraction.
- Built a **custom LLM agent from scratch** (no LangChain or orchestration frameworks) that calls LLM APIs directly to autonomously research and evaluate task decomposition strategies; designed benchmarking pipelines enabling systematic comparison of hierarchical clustering approaches.
- Ran molecular dynamics simulations with **NAMD**, **CHARMM**, and **VMD**.

PROJECTS

SmallC Compiler

March 2026 – April 2026

OCaml

- Built a complete **lexer, recursive descent parser, and evaluator** for a C-like language from scratch in OCaml
- Tokenized raw source via regex, handling operator precedence, keywords, and edge cases; parsed a CFG into an **AST** supporting expressions, control flow, and function definitions with static type checking

Memory Allocator

September 2025 – October 2025

C, Linux, Valgrind, GDB

- Implemented a dynamic memory allocator in C from scratch, manually managing heap space via custom equivalents with explicit block sizing, coalescing, and placement strategies (first-fit/best-fit)

TECHNICAL SKILLS

Languages: Java, C, OCaml, Rust, Python, Assembly, R, Bash, TypeScript

Frameworks & Libraries: Spring Boot, Angular, Spring AI, Ollama, Gemma, GPT APIs

Tools & Platforms: Docker, Git, Azure, GitLab CI/CD, LiteLLM, Elasticsearch, Microsoft Graph API, SharePoint Webhooks

Concepts: Retrieval-Augmented Generation (RAG), Vector Stores, CI/CD Automation, Prompt Engineering